

Pruthviraj Mane

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Robotics | System Modeling and Control | Simulation Engineer

EDUCATION

- **Indian Institute of Technology (IIT), Guwahati** Guwahati, Assam
MS(Research) E-mobility 2020 – 2022
- **Walchand College of Engineering, Sangli** Sangli, Maharashtra
B.Tech in Mechanical Engineering 2015 – 2019

EXPERIENCE

- **Artificial Intelligence and Robotics Laboratory, IISc Bangalore** Bangalore, Karnataka
Research Engineer II working with Dr. Suresh Sundaram, Chair, RBCPS, IISc Oct 2022 - Present
 - **Modeling and Control:**
 - Designed and simulated an underactuated AUV model in MATLAB/Simulink, ensuring stable 6-DOF control and validating behavior in ROS-Gazebo, while exploring cascaded PID, SMC, and MPC controllers; also worked on path planning, obstacle avoidance, and object detection with the same AUV.
 - Integrated IMU, camera, DVL, sonar to collect synthetic data for training navigation models.
 - **Path Planning and Autonomous Navigation:**
 - Developed EROAS, a real-time sonar-guided 3D obstacle-avoidance system for AUVs, by combining a 2.5D sonar sensing approach with a Control Barrier Function (CBF) safety filter to enable safe navigation in cluttered underwater environments.
 - Implemented a physics informed neural network with CBF for efficient obstacle avoidance of AUV.
 - **Simulation Platforms and Synthetic Data Generation:**
 - Built a high-fidelity underwater AUV simulation environment using Unreal Engine 5, enabling photorealistic synthetic visual dataset generation for subsea vision tasks like image enhancement, scene reconstruction, etc.
 - Created a ROS-Gazebo-Unreal ecosystem for generating co-registered datasets and testing navigation algorithms.
 - Developed a maritime ship simulator in Unreal Engine for dynamic behavior modeling and multimodal synthetic data collection as well as testing visual navigation models.
 - **Fabrication and Field Testing of Research ASV:**
 - Designed and built a research ASV with integrated propulsion, sensors, and control hardware.
 - Conducted field trials in lake to validate waypoint navigation, stability, and overall system reliability.

KEY ACADEMIC PROJECTS

- **Lateral dynamics and stability of electric vehicle (Torque Vectoring) — MS Thesis (Jan 2020 – Jul 2022):**
Dr. B. S. Reddy, Assistant Professor, Dept of Mechanical Engineering IITG
Analyzed torque-vectoring techniques for lateral stability of a four-wheel electric vehicle by controlling yaw moment, and modeled comparative controllers—including LQR, MPC, and SMC—to evaluate their effectiveness.
- **Implemented Adaptive Cruise Control (ACC) for Electric Vehicle as an ADAS Level-1 Feature:**
Modeled an ADAS Level-1 adaptive cruise control system in MATLAB/Simulink for an electric vehicle and implemented a PI controller to maintain stable speed when the lead vehicle is beyond the safe distance.

SKILLS

- **Programming and Frameworks:** Python, C/C++, ROS, ROS2, Pytorch, MATLAB, Simulink, docker, git
- **Design and Simulation Tools:** Solidworks (CAD), Gazebo, Unreal Engine 5, Isaac Sim

ACHIEVEMENTS AND RESPONSIBILITIES

- **Authored** three conference papers, one book chapter, and submitted one journal paper for publication as part of research at AIRL, IISc covering areas like intelligent control, safe navigation, etc.
- Led a 30-member interdisciplinary college team in Robocon 2018, overseeing design and execution of the project, resulting in an All-India Rank 38 out of 107 teams.
- Led the Futsal event as Event Head at WEST'19, a national-level sports tournament at WCE Sangli.
- **Ranked top 3 %** among 1913 teams in E-Yantra Hackathon 2020: Fighting COVID-19.
- **AIR 1615** in GATE-ME 2020.